41ST IAHR WORLD CONGRESS SINGAPORE
22 – 27 JUNE 2025
INNOVATIVE WATER ENGINEERING FOR SUSTAINABLE DEVELOPMENT

Book of Extended Abstracts
41st IAHR World Congress
22-27 June 2025
Singapore

SUSTAINABLE WATER TRANSPORT AND LAKE RESTORATION: INTEGRATING AI AND SATELLITE DATA FOR SUKHNA LAKE IN CHANDIGARH GREEN CITY, INDIA FOR ACHIEVING TO SDGs-2030

Badveeti ADINARAYANA[1] AYUSH[2]

[1*]Assistant Professor, Civil Engineering Department, Punjab Engineering College (DU)
Chandigarh-160012
[2]B. Tech Student, Civil Engineering Department, Punjab Engineering College (DU)
Chandigarh-160012
Corresponding Email: badveetiadi@pec.edu.in, badveetiadi@gmail.com,
https://orcid.org/0000-0002-3347-8065
Email: chauhanayush485@gmail.com

ABSTRACT

This paper presents a comprehensive methodology to address the challenges of eutrophication and ecological degradation in Sukhna Lake (Chandigarh, India), combining AI-driven predictive modeling with eco-innovative solutions. Utilizing satellite remote sensing data from Landsat-7 ETM+ and ASTER, alongside in-situ measurements, the proposed framework predicts key water transport quality parameters, such as Secchi Disk Transparency (SDT), Chlorophyll-a, and Land Surface Temperature (LST). These parameters are critical for forecasting trends in eutrophication and assessing overall lake health. The study also introduces floating treatment wetlands and bioremediation as nature-based, adaptive solutions to mitigate nutrient inflow and improve water quality. The AI model provides a proactive, scalable approach for urban lake restoration, offering valuable insights for sustainable ecosystem management practices in urban lakes worldwide. This paper represents a step toward merging cutting-edge technology (AI and satellite remote sensing) with sustainable water transport of ecological interventions to tackle pressing sustainble environmental issues like eutrophication. The goal is not only to restore Sukhna Lake but also to create a scalable framework that can be used globally for urban lake restoration and sustainable transport ecosystem management for Chandigarh city in India.

Keywords: *Sustainable Water Transport, AI, Remote sensing, Land Surface Temperature (LST), Sukhna Lake*